



ERIK ARDIANSYAH

Electrical Engineering

D4 Robotics Engineering Technology

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QR Code Portofolio

Profile

Electrical Engineering graduate (Robotics Engineering Technology) from Batam State Polytechnic with expertise in PLC-based system control design, electrical wiring, and 3D CAD modeling using SolidWorks. Experienced in leading academic engineering projects, compiling technical documentation, and troubleshooting electro-mechanical systems. Ready to contribute to the operation, design, and optimization of industrial automation systems.

Education

Politeknik Negeri Batam | 2022 – 2026

- Bachelor of Applied Science in Robotics Engineering (GPA: 3.40/4.00). Part of the Electrical Engineering Department.

SMK Negeri 3 Pariaman | 2019 – 2022

- Vocational High School Diploma in Computer and Network Engineering (TKJ).

Technical Skills

Robotics & Automation

- PLC Programming
- Industrial Automation Systems
- Electrical Wiring & Troubleshooting
- Sensor Integration
- PID Control Implementation
- Robot Motion Control

Programming

- Python (Basic)
- C++ (Basic)
- Plc (Programming)

Mechanical Design & Engineering

- 3D Modeling
- Mechanical Assembly Design
- Structural Analysis
- Engineering Drawing
- Mechanical Fabrication

Data & Documentation

- Technical Documentation
- Data Analysis
- Bill of Hardware (BOH/BHP) Preparation

Software

- Microsoft Excel
- Microsoft Word
- Microsoft PowerPoint
- SolidWorks
- Arduino IDE

Relevant Projects & Experience

Project Based Learning (PBL) – Delivery Robot | Politeknik Negeri Batam

Mechanical Lead (Semesters 6–7):

- Designed robot mechanical structure using SolidWorks, including 3D modeling and assembly design.
- Conducted structural analysis and simulation to validate mechanical performance before manufacturing.
- Optimized material selection to balance strength, durability, and weight efficiency.
- Led fabrication and assembly processes according to engineering specifications.
- Performed maintenance and troubleshooting during testing and development phases.

Electrical Specialist (Semesters 4–5):

- Prepared Bill of Hardware (BOH/BHP) and calculated power requirements for electrical systems.
- Selected electrical components based on system requirements and operational safety.
- Designed and implemented structured wiring systems for power and signal distribution.
- Conducted electrical testing and troubleshooting to ensure system reliability.

Programming (Semester 3):

- Developed robot motion control algorithms using PID Control methods.
- Integrated control systems with sensors and actuators.
- Performed parameter tuning and response analysis to improve movement stability and accuracy.
- Conducted testing and debugging to ensure reliable robot operation.

Achievements

- **3rd Place:** Appropriate Technology Category – PBL EXPO Politeknik Negeri Batam (2024).
- **3rd Place:** Appropriate Technology Category – KEPRI ENGINEERING EXPO (2023).

Organizational Experience

Kumpulan Anak Seni (KUAS) Polibatam (Arts Student Organization)

Chief Advisory Board (2025)

- Provided strategic guidance for organizational development and management.
- Supported decision-making processes and organizational improvement initiatives.
- Assisted executive members in maintaining organizational effectiveness.

Head of Inventory (2024)

- Managed organizational assets and inventory systems.
- Developed inventory documentation and borrowing procedures.
- Improved asset tracking and operational efficiency.

Internal Division Member – Fine Arts PIC (2023)

- Coordinated fine arts activities and member development programs.
- Facilitated communication between members and organizational management.
- Assisted in organizing events and external competition participation.

Soft skills

- Teamwork
- Leadership
- Effective Communication
- Critical Thinking
- Problem Solving
- Adaptability
- Time Management
- Project Coordination

Languages

- Indonesian — Native
- English — Basic Working Proficiency